RIVER VALLEY HIGH SCHOOL
JC 2
PRELIMINARY EXAMINATION

CANDIDATE NAME													
CENTRE NUMBER	**S**					CLASS	20J	INDEX NUMBER					

H2 BIOLOGY **9744/01**

Paper 1 Multiple Choice **24 September 2021**

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.
Write your name, Centre number, index number on the Answer Sheet in the spaces provided unless this has been done for you.
DO **NOT** WRITE IN ANY BARCODES.

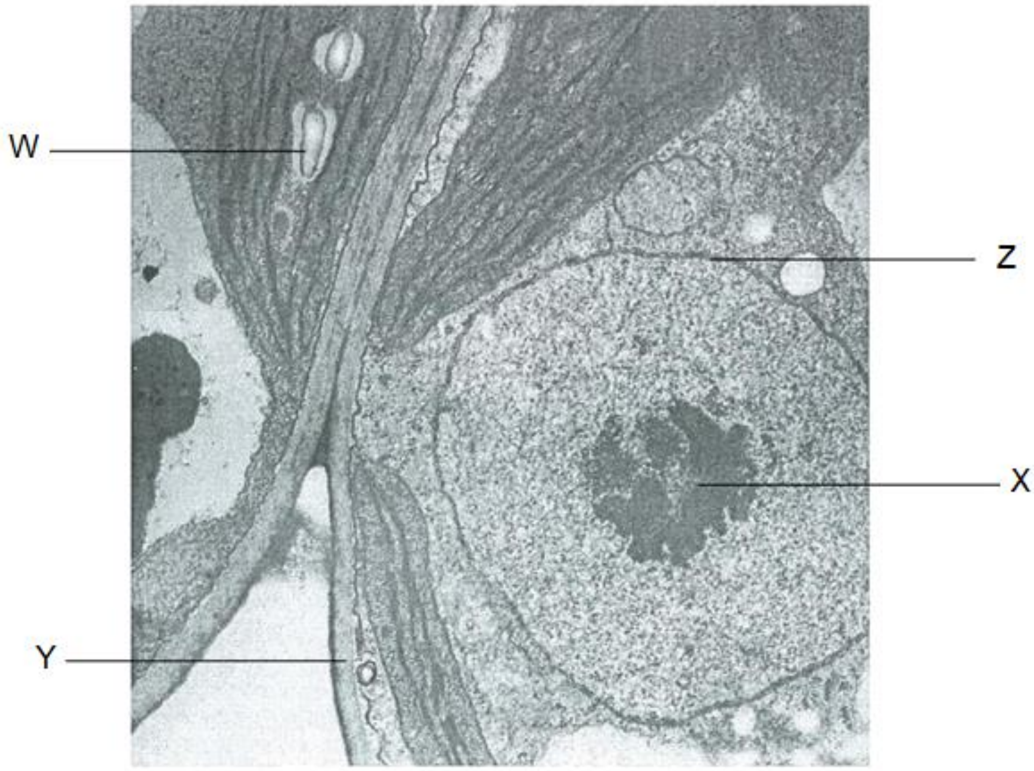
There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C**, and **D**.
Choose the one you consider correct and record your choice **in soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator is expected, where appropriate.

This Question Paper consists of **18** printed pages and **2** blank pages.

1 The electron micrograph shows two adjacent plant cells.



Which row(s) correctly matches the structure to its function?

	structure	function
1	W	manufactures carbohydrates
2	X	manufactures rRNA
3	Y	regulates the movement of substances
4	Z	gives the cell a fixed shape

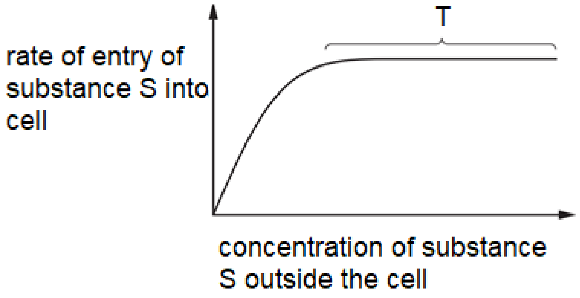
A 2 only

B 3 only

C 1 and 4 only

D 2, 3 and 4 only

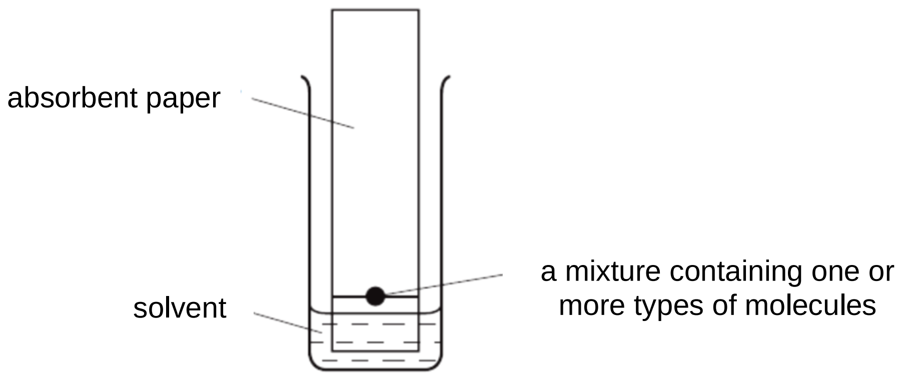
2 The graph shows the rate of movement of S into a cell.



Which row is correct?

	substance S	reason for plateau at T
A	glucose	the cell has used up its supply of ATP
B	glutamic acid	all the carrier proteins are saturated with S
C	oxygen	concentrations of S inside and outside the cell are equal
D	carbon dioxide	concentration of S is no longer a limiting factor

3 Chromatography is a technique used to separate molecules by their solubility. The diagram shows how chromatography is performed.

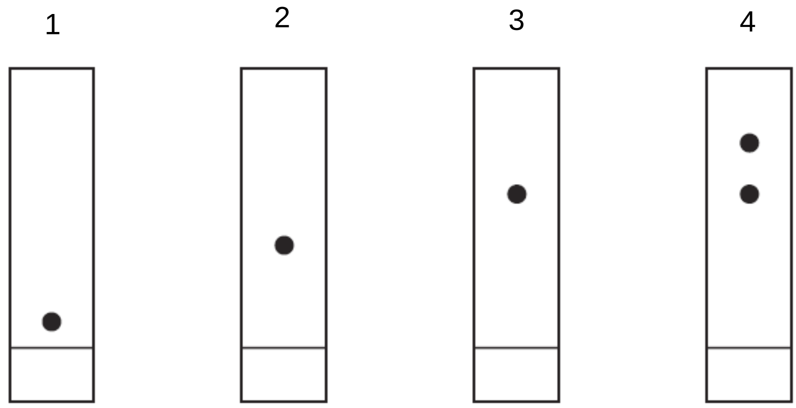


As the solvent rises up the paper, the molecules with the greatest solubility in the solvent travel the greatest distances up the paper. When the solvent reaches the top of the paper, the paper is removed, dried and sprayed with a dye. The different molecules appear as coloured spots.

Chromatography was carried out on four different samples. The samples were:

- sucrose
- cellulose
- products of complete hydrolysis of sucrose
- products of complete hydrolysis of cellulose.

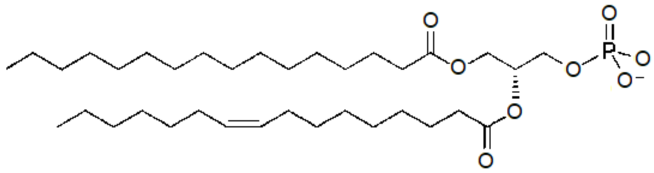
The diagram shows the chromatography results.



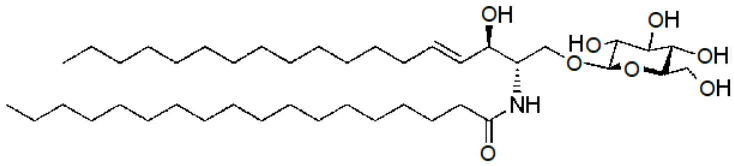
Which row shows the correct results?

	sucrose	cellulose	products of complete hydrolysis of sucrose	products of complete hydrolysis of cellulose
A	1	3	4	2
B	3	1	2	4
C	1	2	3	4
D	2	1	4	3

4 The diagram below shows the structure of two molecules, P and Q.



molecule P



molecule Q

Which statements regarding molecules P and Q are correct?

1 Molecules P and Q are amphipathic.

2 Molecule P is contained within the secretory vesicle.

3 Molecule Q may be involved in blood group determination.

A 1 and 2 only

B 1 and 3 only

C 2 and 3 only

D All of the above

5 Which of these statements are correct?

1 At pH 13, amino acids have a net positive charge.

2 Formation of a peptide bond involves a reaction between an amino group and a carbonyl group.

3 A total of 27 distinct tripeptides can be formed from one valine molecule, one alanine molecule and one leucine molecule.

4 Extreme heating of proteins results in proteins unfolding to secondary structure.

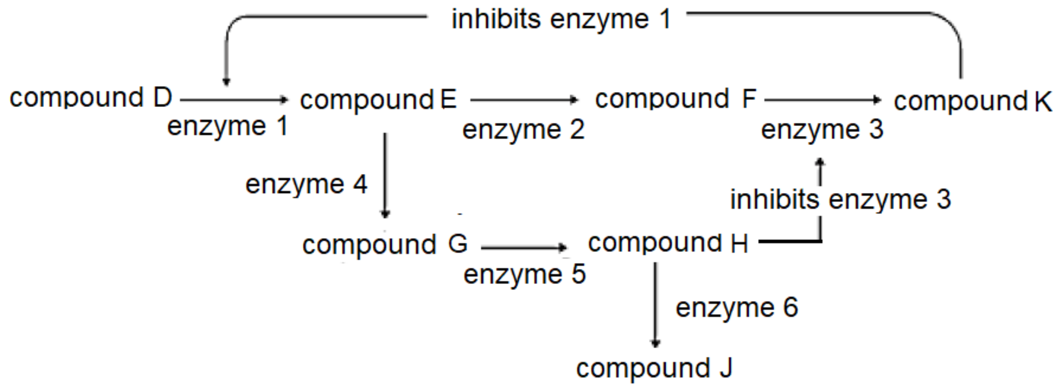
A 2 and 4 only

B 1, 2 and 3 only

C 1, 3 and 4 only

D None of the above

6 The chart below shows the synthesis pathway of compound J and compound K.



Which of the following conclusions are correct?

1. When more enzyme 5 is added, the concentration of all compounds will increase except compound K.
2. When enzyme 6 is removed, there would be more compound F.
3. When enzymes 1, enzyme 2 and enzyme 3 are present, the production of compound K would be continuous if there was a continuous supply of compound D.
4. When all the enzymes are present, the production of compound J would be continuous if there was a continuous supply of compound D.

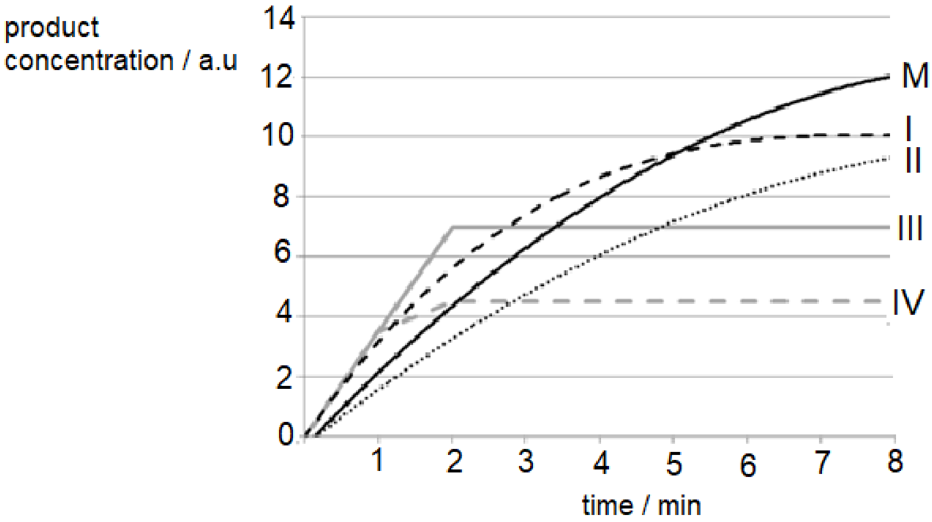
A 1 and 3 only

B 1 and 4 only

C 2 and 3 only

D 2 and 4 only

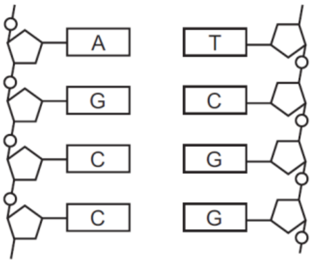
7 Curve M depicts the concentration of product formed by an enzyme-controlled reaction proceeding at 1.0 % substrate concentration, 35 °C, optimum pH of 7.5 and in the absence of any inhibitors.



Which of the conditions stated for curves I to IV is true?

	curve	conditions
A	**I**	1.0% substrate concentration, 45 °C, pH 7.5, non-competitive inhibitor added
B	**II**	1.0% substrate concentration, 45 °C, pH 8.0, competitive inhibitor added
C	**III**	2.0% substrate concentration, 35 °C, pH 14.0
D	**IV**	0.5% substrate concentration, 35 °C, pH 7.5

8 The diagram shows part of a DNA molecule.



Given that pyrimidine bases contain four carbons and purine bases contain five carbons, which row is correct?

	number of carbon atoms in the DNA molecule	number of hydrogen bonds bolding the DNA molecule together
A	36	9
B	36	11
C	76	9
D	76	11

9 The table shows the tRNA anticodons for three amino acids.

amino acid	anticodon (tRNA)
asparagine	5'- AUU -3'
	5'- GUU -3'
glutamic acid	5'- UUC -3'
	5'- CUC -3'
proline	5'- GGG -3'
	5'- AGG -3'

The amino acid sequence of a short segment of a polypeptide is shown below.

C-terminus – glutamic acid – asparagine – asparagine – proline – N-terminus

What was the sequence of bases on the DNA from which this was formed?

A 5'- GGATTATTGCTT -3'

B 5'- TTCGTTATTAGG -3'

C 5'- AGGATTGTTCTC -3'

D 5'- CTCTTGTTAGGA -3'

10 The table shows a comparison of the genomes and protein-coding genes of the prokaryote *Escherichia coli* and the eukaryote *Saccharomyces cerevisiae*.

	E. coli	*S. cerevisiae*
Genome length / base pairs	4 640 000	12 068 000
Number of protein-coding genes	4300	5800
Proteins with roles in:		
metabolism	650	650
energy release	240	175
transcription	240	400
translation	180	350
cell structure	180	250

Which of the following statements account for the differences in the number of protein-coding genes?

1 More catabolic pathways for using carbon compounds in prokaryotes.

2 The presence of introns in eukaryotes.

3 The presence of membrane-bound organelles in eukaryotes.

4 More types of regulatory mechanisms for gene expression in eukaryotes.

A 1 only

B 1 and 2 only

C 1, 3 and 4 only

D 2, 3 and 4 only

11 In an experiment, the 5’ cap and the length of 3’ poly(A) tail of an mRNA are modified.

The table illustrates the effects of capping and polyadenylation on the half-life of the mRNA and the amount of its protein products.

mRNA	mRNA half-life / min	amount of protein / arbitrary units
5’ cap removed and no poly(A) tail	7	940
5’ cap removed with 50-adenine tail	26	3200
5’ cap present but no poly(A) tail	33	10010
5’ cap present with 50-adenine tail	63	280000

Which of the following can be concluded from the results obtained?

1 Capping and polyadenylation have the same effect on the rate of translation of the mRNA.

2 Both capping and polyadenylation are required for the binding of ribosomes.

3 Presence of poly(A) tail results in an increase in amount of protein produced by up to 28 times.

4 Polyadenylation protects against the mRNA degradation more than capping.

A 3 only

B 1 and 2 only

C 3 and 4 only

D 1, 2, 3 and 4

12 Determining the number of CGG repeats can help predict the possibility of developing Huntington disease. The unique sequence outside the repeat region is shown.

5’- ATTCGGACTTG ------(CAG)$_n$------ GTCCAGCTAGAGG -3’

3’- TAAGCCTGAAC ------(GUC)$_n$------- CAGGTCGATCTCC -5’

Which of the following sets of primers can be used in the polymerase chain reaction for the amplification of the repeat region?

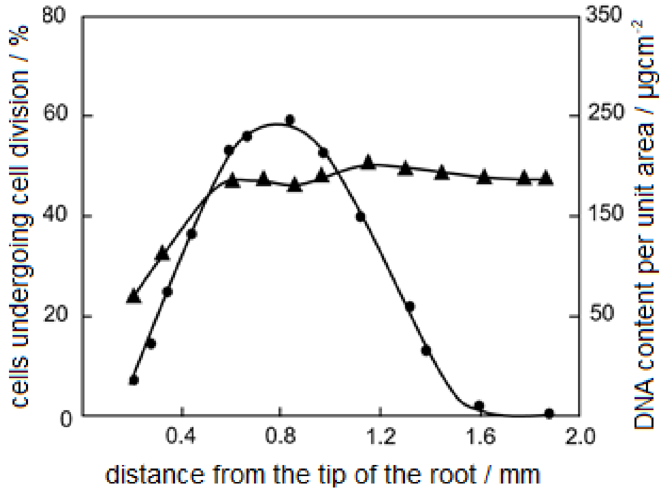
A 5’- CCTCTAG -3’
5’- GTCCAGC -3’

B 5’- ATTCGGA -3’
5’- CCTCTAG -3’

C 5’- GGACTTG -3’
5’- GTCCAGC -3’

D 5’- CCTCTUG -3’
5’- GTCCUGC -3’

13 A student recorded the changes observed along the length of the root of a plant. The graph shows the percentage of cells undergoing cell division and the DNA content per unit area of root tissue at different distances from the tip of the root of the plant.



key ▲ DNA content per unit area ● Percentage of cells undergoing cell division

Which statements correctly explain the graph?

1 Cells only underwent reductive division after 0.2 mm from the root tip.

2 Cells undergoing cell division are smaller in size.

3 Drugs preventing spindle assembly were added to the growing root tip previously.

4 The root is thicker with increasing distance from the root tip.

A 1 and 3 only

B 1 and 4 only

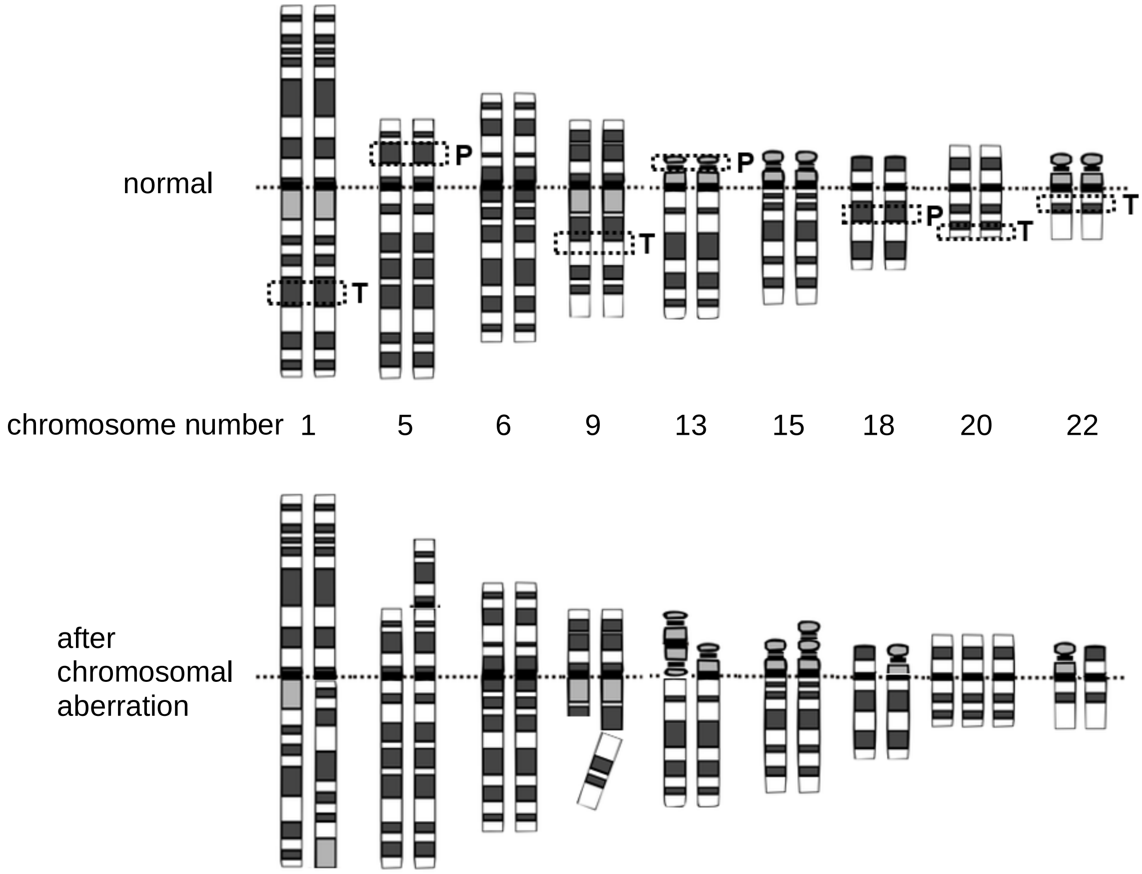
C 2 and 3 only

D 2 and 4 only

14 Which row correctly identifies the stage of meiotic cell cycle with the amount and number of sets of DNA as compared to a liver cell in the same organism?

	compared to liver cell	
	double amount of DNA	same number of sets of DNA
A	prophase I	metaphase II
B	telophase I	anaphase I
C	G2 phase	anaphase II
D	G1 phase	metaphase I

15 A portion of an individual's normal karyotype and after it has undergone chromosomal aberration are shown below. Letters **T** and **P** represent tumour suppressor gene and proto-oncogene respectively and the position of these genes are shown in the "normal" diagram.



What is the number of chromosomal aberrations that may result in cancer development?

A 1

B 2

C 3

D 4

16 Which of the following statements are true about **all** stem cells?

1. Stem cells extracted from any tissue can differentiate into skin cells with the introduction of appropriate transcription factors.
2. Stem cells can develop into whole organism when implanted into the uterus.
3. One potential side effect of any stem cell-based therapy is the formation of a tumour.
4. Stem cells can no longer be extracted once the embryo develops into a foetus.

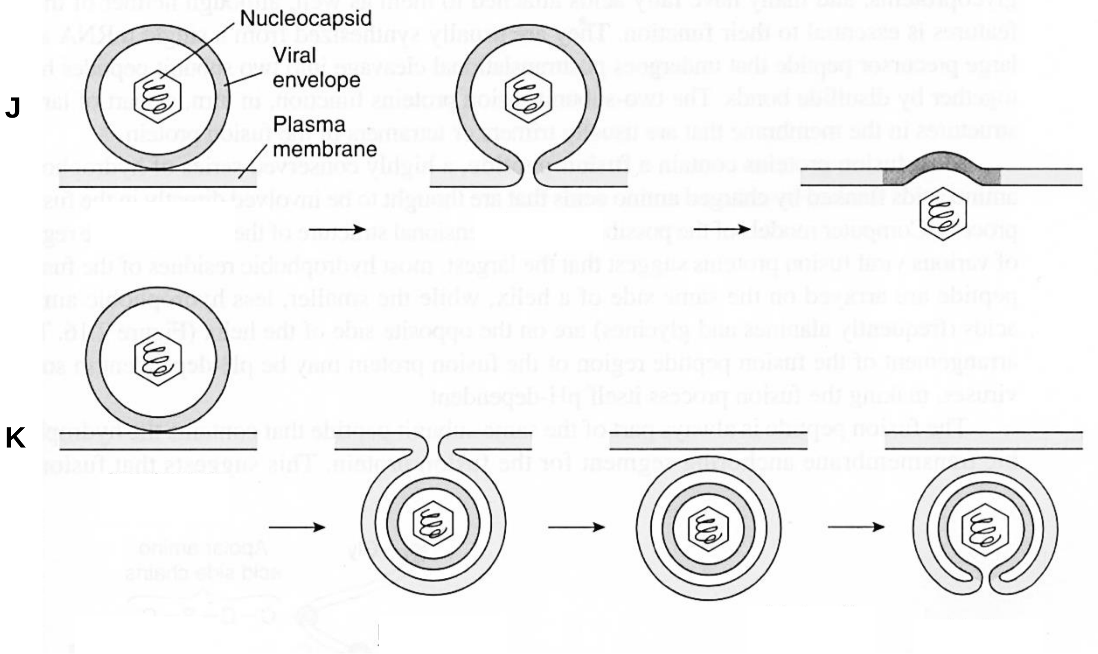
A 1 and 3 only

B 1 and 4 only

C 2 and 3 only

D 3 and 4 only

17 The diagram below shows the entry of two different animal viruses **J** and **K** into its host cell.



Which of the following row is correct?

	J		K	
	identity	present in nucleocapsid	identity	present in nucleocapsid
A	influenza	reverse transcriptase	human immunodeficiency virus	double stranded DNA

B	influenza	single stranded RNA	human immunodeficiency virus	RNA polymerase
C	human immunodeficiency virus	reverse transcriptase	influenza	double stranded DNA
D	human immunodeficiency virus	single stranded RNA	influenza	RNA polymerase

18 Which statement about the prokaryotic and eukaryotic genome is **true**?

A The ratio of promoter to genes is higher in prokaryotic genome than in eukaryotic genome.

B Regulatory sequences are only present upstream of the genes in both prokaryotic and eukaryotic genome.

C Non-histone proteins are involved in condensation of the prokaryotic genome but not the eukaryotic genome.

D Both prokaryotic and eukaryotic genome are made up of double-stranded DNA containing short sequence repeats.

19 A student made the following statements regarding the regulation of *lac* operon.

1 Catabolite activator protein (CAP) is involved in positive regulation of *lac* operon when glucose-CAP complex binds to promoter to recruit RNA polymerase.

2 Mutation of *lac* regulatory gene will always results in constitutive expression of *lac* operon even when lactose is absent.

3 Basal expression of *lac* operon occurs even in the absence of lactose.

Which statements are **not** correct?

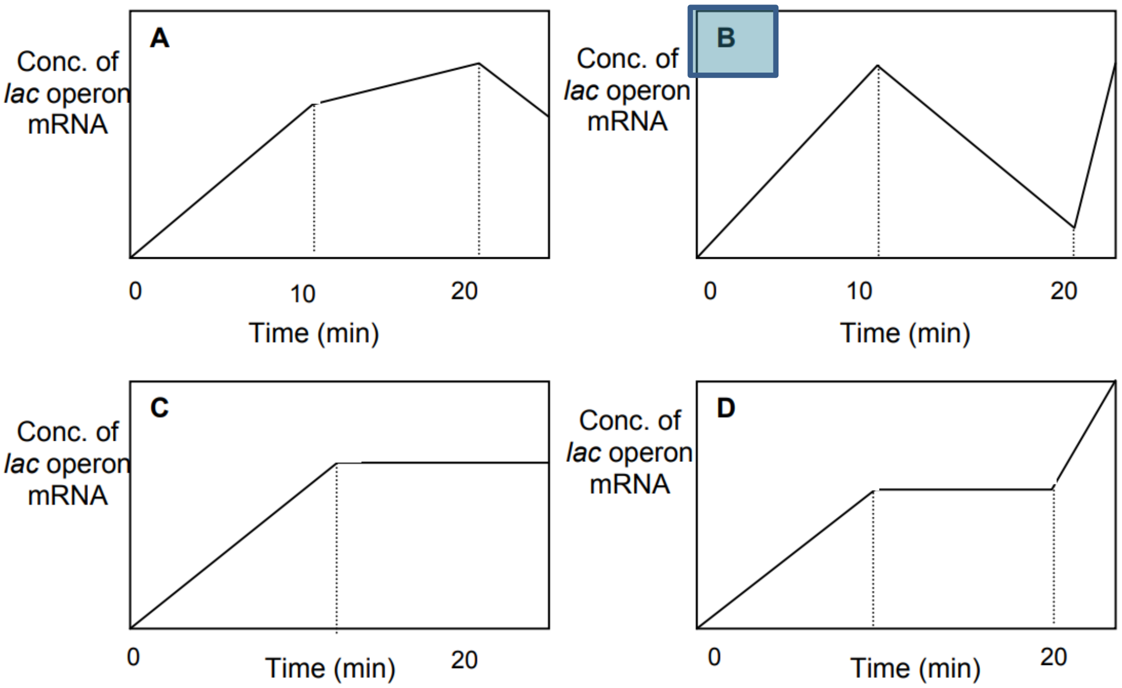
A 1 and 2 only

B 1 and 3 only

C 2 and 3 only

D 1, 2 and 3

20 IPTG is an analogue of allolactose that binds to the lac repressor. A culture of *E. coli* cells, which were grown in the absence of lactose and glucose, were initially supplemented with IPTG. At 10 minutes and at 20 minutes of the experiment, glucose and cAMP were added to the culture respectively. Which of the following graphs represents the amount of *lac* operon mRNA during the time course of this experiment?



21 The inheritance of fruit colour of eggplant is controlled at two different gene loci, E and F.

The colour of eggplant fruit can be either purple or white.

An eggplant, heterozygous at both gene loci, is self-pollinated. Out of 30 offspring, 5 white and 5 purple offspring were randomly sampled. Their genotypes were determined and summarised in the table below.

eggplant fruit colour	genotypes
purple	EeFf, EEFf. EeFF
white	eeFf, eeff, Eeff, EEff

What percentage of the eggplants will have the white phenotype?

A 7%

B 19%

C 44%

D 56%

22 The gene loci W, X, Y and Z forms a linkage group.

The table below shows the recombination frequencies when different pairs of gene loci in this linkage group were investigated.

pairs of gene loci	recombination frequency / %
W and X	46
W and Y	8
X and Y	52
X and Z	13
Y and Z	41

Which of the following shows the correct order of genes on the chromosome?

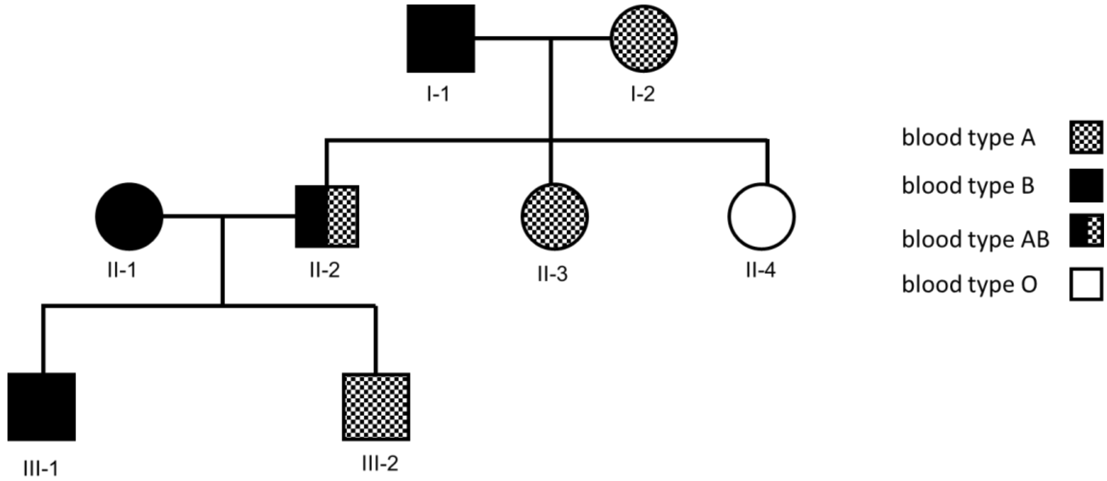
A W, X, Y, Z

B W, Y, Z, X

C Y, W, Z, X

D Y, Z, X, W

23 The pedigree chart below shows the inheritance of blood type in a family.



Which of the following statements are true?

1 I-1 is a homozygote while I-2 is a heterozygote for blood type.
2 If II-4 marries an individual with blood type AB, there is a 25% chance that they will have a daughter with blood type A.
3 The third child of individuals II-1 and II-2 has a 0% chance of being blood type O.
4 The future offspring of individual II-3 will either be blood type A or blood type O.

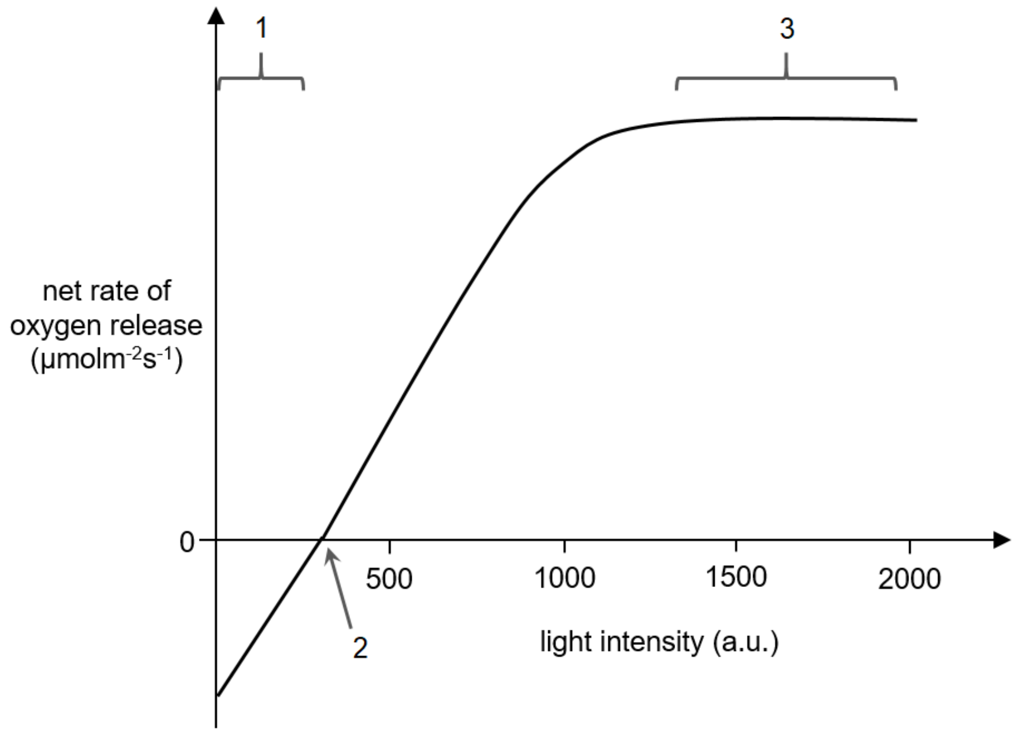
A 1 and 2

B 1 and 3

C 2 and 3

D 3 and 4

24 The graph shows the effect of light intensity on the net rate of oxygen release by a plant.



Which statements about the labelled regions of the graph are consistent with the results?

	1	2	3
A	net loss of mass occurs	rate of ATP production in respiration is equal to rate of ATP used in photosynthesis	more RuBP regenerated when temperature is increased
B	net gain of mass occurs	rate of carbon dioxide production in respiration is equal to rate of carbon dioxide used in photosynthesis	less water lysed when temperature is increased
C	net loss of mass occurs	rate of carbon dioxide production in respiration is equal to rate of carbon dioxide used in photosynthesis	less NADH produced when temperature is decreased
D	net gain of mass occurs	rate of ATP production in respiration is equal to rate of ATP used in photosynthesis	more NADPH is produced when temperature is increased

25 Which statement about ATP synthesis is **correct**?

A In animal cells, a molecule of lactate can yield 15 ATP upon complete oxidation.

B Reduced NADP and reduced FAD carries energised electrons which can be used to generate a proton gradient across the inner mitochondrial membrane.

C In plant cells, ATP synthesis only occurs in the chloroplasts and mitochondria.

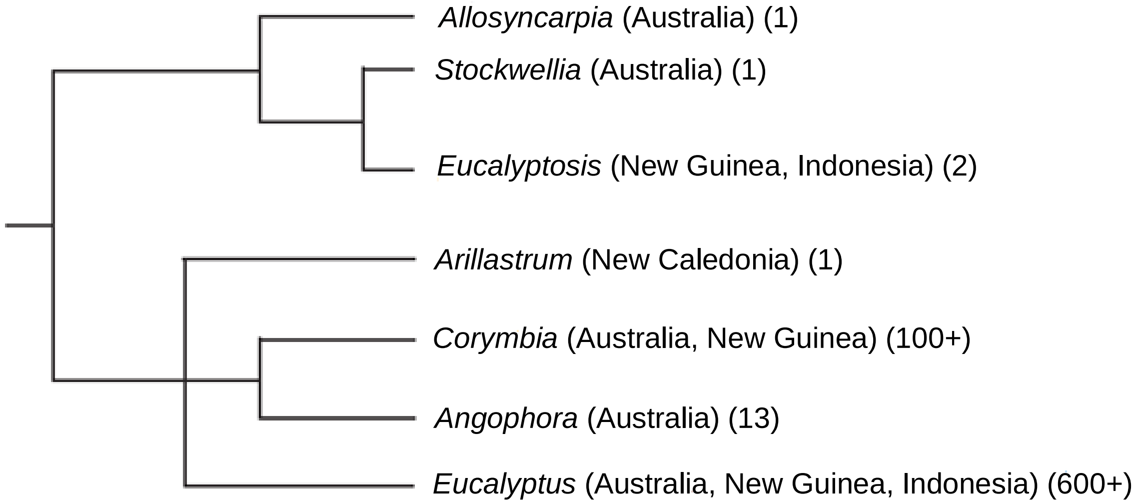
D For every molecule of glucose, one ATP is produced during Krebs cycle.

26 Which row shows the difference between **all** receptor tyrosine kinase and G-protein linked receptor?

	receptor tyrosine kinase	G-protein linked receptor
A	ligand must be water soluble	ligand can be water soluble or fat soluble
B	activation involves dimerisation of receptor subunits	activation involves conformation change
C	activation involves ATP	activation involves GTP
D	does not involve second messengers	involves second messengers

27 Eucalypts are commonly known as gum trees and they originate from Australia. They have been recently classified into seven different genera on the basis of nuclear and chloroplast DNA sequencing.

A proposed phylogeny for the seven genera is shown in the diagram, along with the countries in which they are found and the number of species in the genus.



Which statement(s) regarding the Eucalypts can be concluded from the diagram?

1 *Allosyncarpia* is more closely related to *Eucalyptosis* than *Arillastrum*.

2 *Eucalyptus* is the fittest amongst all the genera of Eucalypts in Australia.

3 Sympatric speciation resulted in the different genera in New Guinea.

A 1 only

B 1 and 2 only

C 2 and 3 only

D 1, 2 and 3

28 Researchers examined the effect of wood smoke in early humans and found the following.

- *AHR* gene codes for a protein that binds to toxic substances, such as polycyclic aromatic hydrocarbons (PAH) from wood smoke. This protein complex enters the nucleus to activate genes that result in toxic effects and even cell death.
- In non-human primates and mice, the AHR protein has an alanine at position 381.
- In humans and our closest extinct relative, the Neanderthal, the AHR protein has a single amino acid change at position 381 from alanine to valine. This reduces the ability of AHR to bind PAH.
- Around 800 000 years ago, after humans split from the lineage with the Neanderthals, this mutation rose to a high frequency in the human lineage.
- The increase in allele frequency of the mutated gene coincided with the ability of early humans to harness fire for various purposes compared to early Neanderthals.

What may be concluded from the findings?

1 Mutation is an important driver of macro-evolution.
2 Different environmental niches confer different selective advantages.
3 Lack of gene flow causes the mutated *AHR* gene to accumulate in the humans.
4 Wood was the main fuel source for early humans.

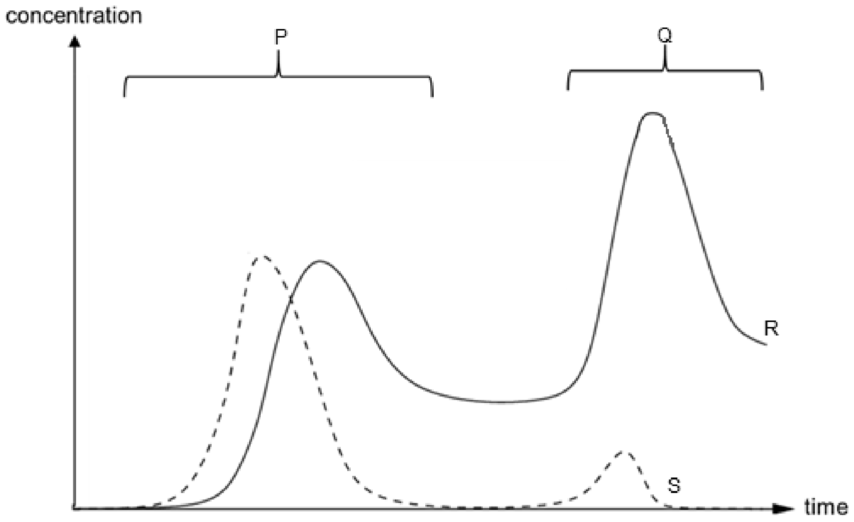
A 1 and 3 only

B 1 and 4 only

C 2 and 3 only

D 2 and 4 only

29 The graph below shows the changes in concentration of viral mRNA and cytotoxic T cell in the blood of an individual.



Which row correctly matches P, Q, R and S?

	P	Q	R	S
A	primary immune response	secondary immune response	cytotoxic T cell	viral mRNA
B	innate immune response	specific immune response	viral mRNA	cytotoxic T cell
C	primary immune response	secondary immune response	viral mRNA	cytotoxic T cell
D	innate immune response	specific immune response	cytotoxic T cell	viral mRNA

~~**30**~~ ~~Endemic species are defined as any species that live in a limited geographical area. They are especially affected by climate change.~~

~~Which reasons explain this?~~

~~1 They have reduced genetic variation.~~

~~2 They are highly adapted to their habitat.~~

~~3 They are outcompeted by invasive species.~~

~~4 They have a smaller population size.~~

~~**A**~~ ~~1 and 2 only~~

~~**B**~~ ~~2 and 3 only~~

~~**C**~~ ~~1, 3 and 4 only~~

~~**D**~~ ~~1, 2, 3 and 4~~

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